



Geotech

GEO-JPMG(DSV) Proton Magnetometer



Introduction

In nature, the magnetism of rocks, ores, and geological structures is uneven. In the geomagnetic field, this magnetic inhomogeneity causes changes in the spatial distribution of the geomagnetic field. When the magnetic field of magnetic bodies is superimposed on the normal geomagnetic field, the resulting superimposed magnetic field is called a magnetic anomaly. By detecting magnetic anomalies with a proton precession magnetometer, the distribution of magnetic ore bodies and geological structures can be inferred.

Geotech has spent two years developing a new - generation proton magnetometer. Compared with other proton magnetometers, it stands out as the fastest and most comprehensive ground - based proton magnetometer to date. It is lightweight and portable. Equipped with a high - accuracy and highly stable oven - controlled crystal oscillator (OCXO), it ensures stable operation and accurate data.

The standard GPS of this magnetometer supports multiple satellite navigation systems including GNSS, GPS, Beidou, and GLONASS. It not only provides precise positioning but also enables time synchronization with an accuracy of up to 30 milliseconds, bringing high efficiency and convenience to field operations.

Moreover, it features three working modes: mobile, walking, and base - station mode, meeting the needs of different scenarios. Whether it is mineral exploration (such as for iron ore, lead - zinc ore, copper ore, etc.), geophysical exploration of geological structures, archaeological excavations, earthquake monitoring, volca-



Applications

- ➔ Engineering investigation, such as pipeline detection, etc
- ➔ Earthquake precursor monitoring, volcano observation and other environmental and disaster geological work
- ➔ Oil and natural gas exploration, research on geological structure and geotectonics related to oil and gas
- ➔ Mineral exploration, such as iron ore, lead-zinc ore, copper ore, etc
- ➔ Base station magnetometer for aerial and marine magnetic survey
- ➔ Detection of small ferromagnetic objects, etc
- ➔ Fault location
- ➔ Archaeology
- ➔ Hydrology



- ➡ It can measure the total field of the geomagnetic field or the gradient measurement (horizontal gradient or vertical gradient, need to add probe and probe holder)
- ➡ Using walking MAG acquisition mode can collect magnetic field data while walking in the field, so as to obtain continuous observation profiles
- ➡ External GNSS, no need to configure baudrate. It can be used to store the coordinate values of the measuring points, and at the same time, the proton magnetometer is time synchronized, and the time synchronization accuracy reaches 30 microseconds
- ➡ Built-in high-precision constant temperature crystal oscillator not only ensures measurement accuracy, but also ensures measurement stability
- ➡ It can display the magnetic field strength curve, easy to operate and intuitive data
- ➡ Can move single point measurement and move continuous measurement
- ➡ Light and portable, the whole system uses a backpack strap, and one person can complete all measurement tasks
- ➡ With USB flash disk transmission and RS-232 transmission, two data transmission modes
- ➡ Professional geological software can draw contour maps, profiles maps, etc.
- ➡ 32MB memory, support modifying



- ➡ Hard aluminum alloy shell, sand blasting and oxidation treatment, ITT imported socket, which can be plugged in and out for tens of thousands of times, also suitable for harsh environments, IP67 grade
- ➡ There is high-precision time synchronization between base station observation and field mobile observation, which is very important to ensure high-precision observation results in magnetic interference area
- ➡ Both full range automatic tuning and manual tuning
- ➡ The signal quality shall be monitored in time, and the degradation of signal quality can be found in time to take remedial measures



Specifications

Sensitivity :	0.05nT
Dynamic range :	20,000-120,000nT
Absolute accuracy :	± 0.1nT
Resolution :	0.01nT
Sample :	3~ 60s @Mobile Mode; 3~3600s @Base Mode.
Gradient Tolerance:	>5000nT/m
Remote Control :	optional remote control via RS-232 interface
Input/Output :	RS-232 via 10 pin weatherproof connector with USB adaptor
Internal Memory :	32MB (209715 readings in Manual Mode, 699050 readings in Base Mode)
Dimension & weight:	Console: 220mm×70mm×175mm, 1.57kg Sensors: 140mm(L)×70mm(D); 1kg
Operating Temperature:	- 40°C to + 55°C

Standard Configuration

GEO-JPMG Console	1x
Harness / Charger	1x
Sensor with Cable	2x
Aluminum rod 50cm	4x
Standard GPS	1x
RS-232 Cable with USB adapter	1x
Instruction Manual	1x
Shipping Case	1x